

The US opiate crisis: poverty, prescription rates and drug-related mortality at the county level

DATA5207 Group Project 2

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May 2026

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Introduction

The United States is in the midst of a sustained drug-overdose epidemic that has, since the late 1990s, killed several hundred thousand people. A central feature of the crisis is its uneven geography: drug-related mortality is concentrated in particular counties, particularly outside of large metropolitan areas, and is closely linked to the supply of prescription opioids and to underlying social and economic disadvantage. This report uses 2016 county-level data from the *US Census Bureau* and the *Centers for Disease Control and Prevention (CDC)* to examine how county poverty rates and opioid prescribing rates jointly predict mortality from drugs and alcohol. We ask: **after accounting for other county-level characteristics, are prescribing rates and poverty independently associated with drug- and alcohol-related mortality?**

Theory

Two complementary literatures motivate the analysis. The first, often called the *deaths of despair* hypothesis (Case and Deaton 2015, 2017), argues that declining economic opportunity in non-urban America has produced excess mortality from suicide, alcohol, and drug overdose among working-age adults. Monnat 2019’s findings support this argument by highlighting an association between economic distress and mortality during this time period, thus we can use county-level poverty acts as a proxy for the structural disadvantage that drives use of, and harm from, drugs and alcohol. We therefore expect poverty to be positively associated with drug- and alcohol-related mortality.

The second is a *supply-side* argument: liberal opioid prescribing conditions on the healthcare end (Owens 2020), especially in the late 1990s and 2000s, created a population of dependent users who later transitioned to illicit opioids when prescription supply tightened (Quinones 2015; Alpert et al. 2018). On this account, the prescribing rate itself is a leading indicator of overdose mortality, independent of underlying poverty. We therefore expect prescribing rates to be positively associated with drug and alcohol mortality, and to remain so after controlling for poverty.

These accounts are not mutually exclusive. The point of the regression analysis below is to estimate each association while holding the other constant, alongside conventional controls for the urban-rural gradient, labour market conditions and education.

Data and methods

Data sources

We combine four county-level datasets:

1. **Poverty.** *American Community Survey 5-year estimates 2016*, table S1702 (US Census Bureau). We use the percentage of families with income below the federal poverty line.
2. **Drug- and alcohol-related mortality.** *CDC WONDER Multiple Cause of Death*, 1999-2016. We use the crude death rate per 100,000 population, as well as the underlying death and population counts for the count-data robustness model.
3. **Opioid prescribing rates.** *CDC US Opioid Dispensing Rate Maps*. We use 2016 prescriptions per 100 persons.
4. **Other county-level predictors.** A consolidated dataset distributed with this assignment (Wu, *COVID-19 US County-level Summaries*; codebook at https://github.com/JieYingWu/COVID-19_US_County-level_Summaries). We use four additional variables: the USDA Rural-Urban Continuum Code (2013), the 2018 unemployment rate, the percentage of adults aged 25+ with a bachelor’s degree or higher (2014-18), and population density.

For the spatial visualisation we use the *US Census Bureau* 2016 cartographic-boundary shapefile of US counties (resolution 1:20m).

Loading and preparing the data

We begin by loading the poverty dataset. The raw file has two header rows, so we skip the first when calling `read.csv()`:

```
poverty.data <- read.csv(
  "Data/US data for group task/poverty/poverty data - ACS_16_5YR_S1702_with_ann.csv",
  skip = 1, header = TRUE
)
```

The file contains 603 columns and 3142 rows – one row per US county. Next we load the CDC mortality file. It is tab-separated, so we use `read.table()`:

```
drugs.mort.data <- read.table(
  "Data/US data for group task/mortality/Drugs and alcohol Cause of Death, 1999-2016.txt",
  sep = "\t", header = TRUE
)
```

The mortality file has 9 variables and 2934 rows. The two files use different names for the county FIPS code (`Id2` in the poverty file, `County.Code` in the mortality file), so we rename one before merging. We also strip the suffixes “County”, “Parish”, “city”, “Borough” and “Census Area” from county names so that the names match those in the shapefile we will use later for mapping. We retain `Deaths` and `Population` from the mortality file because we will need them for the negative-binomial robustness check below.

```
library(tidyverse)

patterns <- c(" County| Parish| city| Borough| Census Area")

county.data <- merge(
  poverty.data %>%
    dplyr::rename(below.poverty.rate =
      All.families....Percent.below.poverty.level..Estimate..Families) %>%
    dplyr::select(Id2, below.poverty.rate),
  drugs.mort.data %>%
    dplyr::rename(Id2 = County.Code,
      mortality.rate = Crude.Rate) %>%
    mutate(county = gsub(patterns, "", County)) %>%
    dplyr::select(Id2, county, mortality.rate, Deaths, Population)
)
```

The merge produces 2928 county-year observations on 6 variables. The CDC suppresses the crude rate when fewer than 20 deaths are recorded (these are coded as `Unreliable`), which both creates missingness and forces the variable to be read as a string. We recode these as `NA` and convert the column to numeric:

```

county.data <- county.data %>%
  mutate(mortality.rate = ifelse(
    mortality.rate == "Unreliable",
    NA,
    suppressWarnings(as.numeric(as.character(mortality.rate)))
  ))

```

After this step 275 counties have a suppressed (missing) mortality rate. We retain them for the count-data robustness model below, which uses raw Deaths and Population rather than the suppressed crude rate.

Adding opioid prescribing rates

We then load the 2016 county-level opioid prescribing file and merge it on the FIPS code, standardising both the poverty rate and the prescribing rate into z-scores so the regression coefficients are directly comparable:

```

opioids.data <- read.csv(
  "Data/US data for group task/opioids/opioids prescribing rates.csv"
)

county.data <- merge(
  county.data %>%
    mutate(z.poverty = (below.poverty.rate - mean(below.poverty.rate, na.rm = TRUE)) /
      sd(below.poverty.rate, na.rm = TRUE)),
  opioids.data %>%
    dplyr::rename(Id2 = FIPS.County.Code) %>%
    mutate(z.opiates = (X2016.Prescribing.Rate -
      mean(X2016.Prescribing.Rate, na.rm = TRUE)) /
      sd(X2016.Prescribing.Rate, na.rm = TRUE)) %>%
    dplyr::select(Id2, z.opiates, X2016.Prescribing.Rate)
)

```

Adding control variables from the consolidated county-level dataset

The consolidated county.data.RData file contains hundreds of variables. Because the object inside it happens to be called county.data – the same name as our merged data frame – we load it into a separate environment and extract the four control variables we want before discarding the rest. We keep the variable names short and informative:

```

controls.env <- new.env()
load(
  "Data/US data for group task/Other census data/county.data.RData",
  envir = controls.env
)

controls <- controls.env$county.data %>%

```

```

dplyr::transmute(
  Id2 = as.integer(FIPS),
  rural.urban.code = as.numeric(Rural.urban.Continuum.Code_2013),
  unemployment.rate = as.numeric(Unemployment_rate_2018),
  pct.bachelors = as.numeric(
    Percent.of.adults.with.a.bachelor.s.degree.or.higher.2014.18
  ),
  pop.density = as.numeric(
    Density.per.square.mile.of.land.area...Population
  )
)

rm(controls.env)

county.data <- merge(county.data, controls, by = "Id2", all.x = TRUE)

```

The Rural-Urban Continuum Code (USDA, 2013) ranges from 1 (most-urban metropolitan counties of one million or more) to 9 (the most rural, non-adjacent to a metropolitan area). Higher values therefore indicate more rural counties.

Loading the US county-level shapefile

For the spatial visualisation we load the Census Bureau's 2016 county-boundary shapefile and the FIPS concordance file, attach state abbreviations, fix three known county-name discrepancies, and join in the analysis variables. Because a small number of county-name strings (e.g. "Bedford city" in Virginia) are duplicated across states, we set `relationship = "many-to-many"` to silence the corresponding warning:

```

library(sf)

us.shape.1 <- st_read(
  "Data/US data for group task/US shapefiles/cb_2016_us_county_20m.shp",
  quiet = TRUE
)

fip.concordance <- read.csv(
  "Data/US data for group task/fips concordance.csv"
)

us.shape.2 <- us.shape.1 %>%
  merge(
    fip.concordance %>%
      mutate(STATEFP = ifelse(
        Numeric.code >= 0 & Numeric.code <= 9,
        paste0(0, Numeric.code), as.character(Numeric.code)
      )) %>%
    dplyr::rename(state = Alpha.code)
  )

us.shape.2$county <- paste0(us.shape.2$NAME, ", ", us.shape.2$state)
us.shape.2$county <- gsub("Doña Ana, NM", "Dona Ana, NM", us.shape.2$county)

```

```

us.shape.2$county <- gsub("LaSalle, LA", "La Salle, LA", us.shape.2$county)
us.shape.2$county <- gsub("Oglala Lakota, SD", "Oglala, SD", us.shape.2$county)

us.shape.2 <- dplyr::left_join(
  us.shape.2,
  county.data %>%
    dplyr::select(county, below.poverty.rate, mortality.rate,
                  X2016.Prescribing.Rate),
  by = "county",
  relationship = "many-to-many"
)

```

Descriptive analysis

Before fitting any regression, we examine the distribution of each variable and the bivariate associations among them. The table below reports summary statistics for the analytic sample (counties with non-missing mortality and prescribing data):

```

analysis.df <- county.data %>%
  dplyr::filter(
    !is.na(mortality.rate),
    !is.na(below.poverty.rate),
    !is.na(X2016.Prescribing.Rate)
  )

# Build the summary table manually so we don't rely on a regex that has
# to cope with parentheses / per-cent signs in column labels.
summary.vars <- list(
  "Drug/alcohol mortality (per 100k)" = analysis.df$mortality.rate,
  "Families below poverty line (%)"   = analysis.df$below.poverty.rate,
  "Opioid Rx per 100 persons (2016)"  = analysis.df$X2016.Prescribing.Rate,
  "Unemployment rate (%)"             = analysis.df$unemployment.rate,
  "Adults with bachelor's degree+ (%)" = analysis.df$pct.bachelors,
  "Rural-urban continuum code (1-9)"  = analysis.df$rural.urban.code
)

summary.tbl <- do.call(rbind, lapply(names(summary.vars), function(nm) {
  x <- summary.vars[[nm]]
  data.frame(
    Variable = nm,
    Mean     = round(mean(x, na.rm = TRUE), 2),
    SD       = round(sd(x, na.rm = TRUE), 2),
    Min      = round(min(x, na.rm = TRUE), 2),
    Median   = round(median(x, na.rm = TRUE), 2),
    Max      = round(max(x, na.rm = TRUE), 2),
    N        = sum(!is.na(x)),
    check.names = FALSE
  )
}))

```

```
knitr::kable(
  summary.tbl,
  row.names = FALSE,
  caption = "Summary statistics for the analytic sample"
)
```

Table 1: Summary statistics for the analytic sample

Variable	Mean	SD	Min	Median	Max	N
Drug/alcohol mortality (per 100k)	20.65	9.33	4.2	18.6	113.4	2612
Families below poverty line (%)	12.18	5.38	1.2	11.5	39.9	2612
Opioid Rx per 100 persons (2016)	80.18	40.48	0.0	76.0	470.3	2612
Unemployment rate (%)	4.20	1.35	1.6	4.0	18.1	2612
Adults with bachelor's degree+ (%)	22.03	9.89	5.4	19.4	78.5	2612
Rural-urban continuum code (1-9)	4.45	2.48	1.0	4.0	9.0	2612

We then plot the distribution of the dependent variable and of each continuous predictor. Drug- and alcohol-related mortality is bounded below at zero and right-skewed – the median county has a rate around 18 per 100,000, but the upper tail extends past 100. A log transformation makes the distribution approximately symmetric, which is the standard treatment for skewed positive rates and is the form we will use as the dependent variable in the OLS models below.

```
library(ggplot2)
library(scales)

p1 <- ggplot(analysis.df, aes(x = mortality.rate)) +
  geom_histogram(bins = 40, fill = "grey40", colour = "white") +
  labs(x = "Drug/alcohol mortality (per 100,000)", y = "Counties") +
  theme_minimal()

p2 <- ggplot(analysis.df, aes(x = mortality.rate)) +
  geom_histogram(bins = 40, fill = "grey40", colour = "white") +
  scale_x_log10() +
  labs(x = "Mortality (log scale)", y = "Counties") +
  theme_minimal()

gridExtra::grid.arrange(p1, p2, ncol = 2)
```

Bivariate scatter plots show the expected positive association of mortality with both poverty and opioid prescribing rates, but neither relationship is strong on its own. Loess smoothers (red) suggest that the association with prescribing rate is closer to linear in the log scale than in the raw scale, supporting our choice of `log(mortality.rate)` as the OLS outcome.

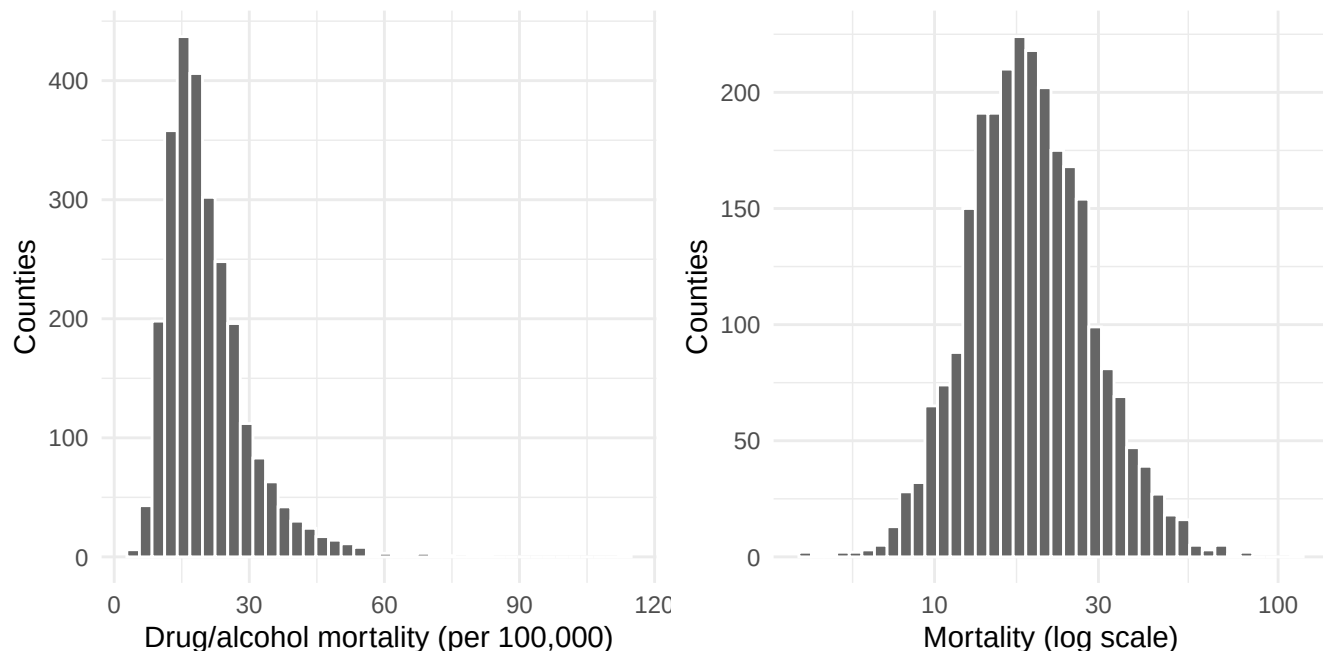


Figure 1: Distribution of drug- and alcohol-related mortality (raw and log scale)

```
s1 <- ggplot(analysis.df,
             aes(x = below.poverty.rate, y = mortality.rate)) +
  geom_point(alpha = 0.25, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, colour = "firebrick") +
  scale_y_log10() +
  labs(x = "Families below poverty (%)",
       y = "Mortality (per 100k, log)") +
  theme_minimal()

s2 <- ggplot(analysis.df,
             aes(x = X2016.Prescribing.Rate, y = mortality.rate)) +
  geom_point(alpha = 0.25, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, colour = "firebrick") +
  scale_y_log10() +
  labs(x = "Opioid Rx per 100 persons (2016)",
       y = "Mortality (per 100k, log)") +
  theme_minimal()

gridExtra::grid.arrange(s1, s2, nrow=1, ncol = 2)
```

When we try to see the relationship between the mortality rate due to drug addiction to unemployment rate of each region and we try to see if most of the deaths are in bachelors or not. The scatterplots shown below show that there is a non-linear relationship between unemployment rates and drug deaths, with the most concentration of drug death rates in regions with about 3-5% unemployment rates and then it starts to become more random. We do see a much more random trend in comparing the percentage of bachelors to deaths due to drugs, here we find that most regions with about 20% bachelor concentration has a higher mortality rate but it decreases as the bachelor concentration increases, which introduces an intriguing question of whether drug deaths are more involved in married individuals or bachelors.

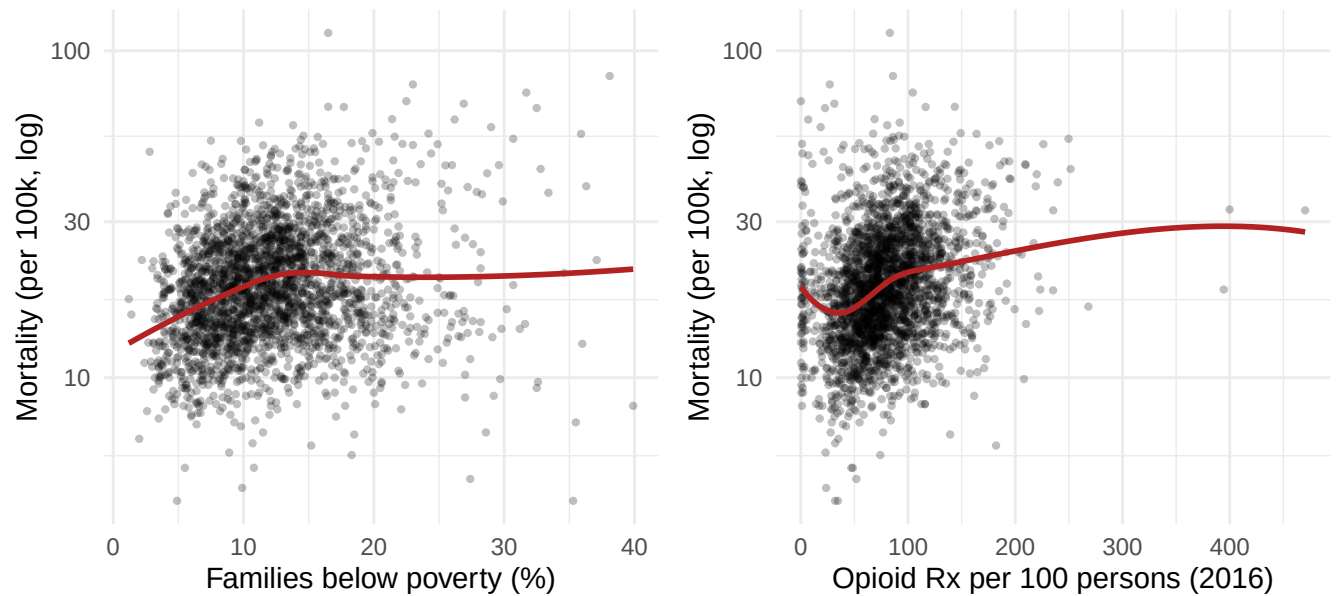


Figure 2: Drug/alcohol mortality vs poverty and opioid prescribing rate (log scale)

```
s3 <- ggplot(analysis.df,
             aes(x = unemployment.rate, y = mortality.rate)) +
  geom_point(alpha = 0.25, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, colour = "firebrick") +
  scale_y_log10() +
  labs(x = "Unemployment Rate (2016)",
       y = "Mortality (per 100k, log)") +
  theme_minimal()
s4 <- ggplot(analysis.df,
             aes(x = pct.bachelors, y = mortality.rate)) +
  geom_point(alpha = 0.25, size = 0.8) +
  geom_smooth(method = "loess", se = FALSE, colour = "firebrick") +
  scale_y_log10() +
  labs(x = "Percent of bachelors (2016)",
       y = "Mortality (per 100k, log)") +
  theme_minimal()
gridExtra::grid.arrange(s3, s4, nrow=1, ncol = 2)
```

Finally we report the pairwise Pearson correlations among the analysis variables (computed pairwise to handle missingness in the controls):

```
cor.mat <- analysis.df %>%
  dplyr::transmute(
    log.mortality = log(mortality.rate),
    poverty       = below.poverty.rate,
    opioid.rx     = X2016.Prescribing.Rate,
    unemployment  = unemployment.rate,
```

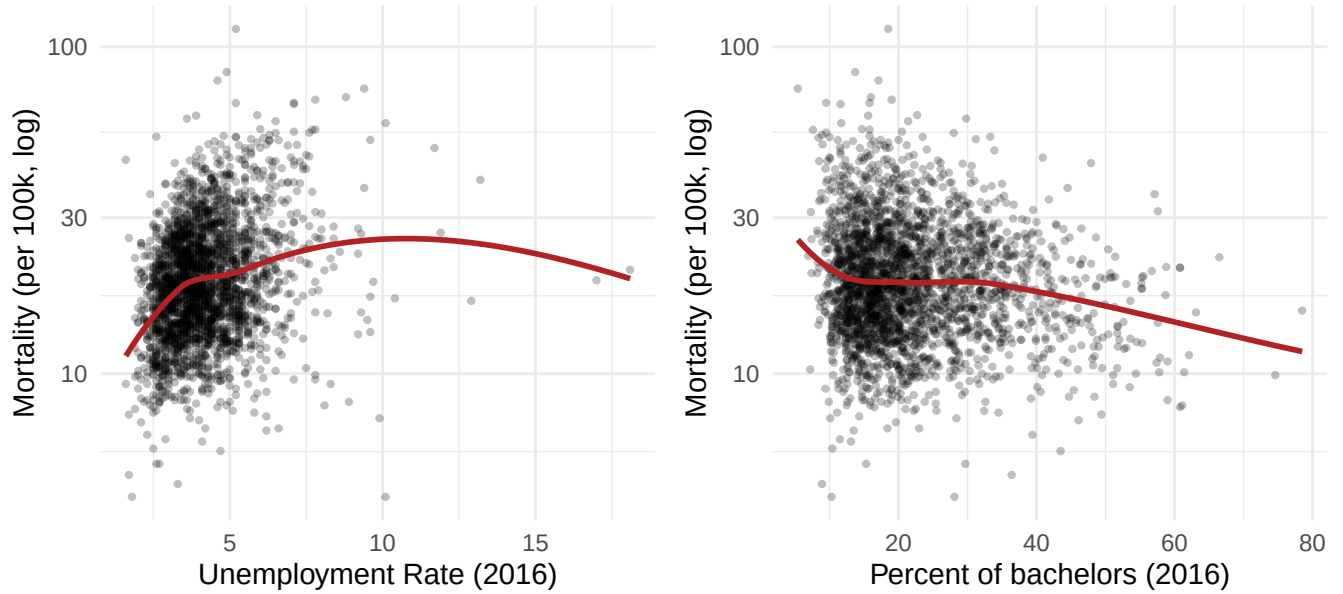


Figure 3: Drug/alcohol mortality vs unemployment rate (log scale)

```
pct.bachelors      = pct.bachelors,
rural.urban.code   = rural.urban.code
) %>%
cor(use = "pairwise.complete.obs") %>%
round(2)

knitr::kable(
  cor.mat,
  caption = "Pairwise Pearson correlations among analysis variables"
)
```

Table 2: Pairwise Pearson correlations among analysis variables

	log.mortality	poverty	opioid.rx	unemployment	pct.bachelors	rural.urban.code
log.mortality	1.00	0.19	0.27	0.26	-0.12	0.08
poverty	0.19	1.00	0.32	0.52	-0.52	0.28
opioid.rx	0.27	0.32	1.00	0.15	-0.21	0.06
unemployment	0.26	0.52	0.15	1.00	-0.38	0.21
pct.bachelors	-0.12	-0.52	-0.21	-0.38	1.00	-0.45
rural.urban.code	0.08	0.28	0.06	0.21	-0.45	1.00

The correlations behave as theory predicts: log-mortality is positively correlated with poverty, opioid prescribing, unemployment, and the rural-urban continuum (i.e. mortality is higher in more rural counties), and negatively correlated with the share of college-educated adults. None of the predictor pairs is so highly correlated as to threaten the multivariate model with collinearity (the largest absolute correlation among the regressors is between unemployment and poverty, around $|r| \approx 0.6$).

Regression specification

We model county-level mortality with three nested ordinary-least-squares specifications:

- **Model 1.** $\log(\text{mortality}_i) = \alpha + \beta_1 z\text{-poverty}_i + \varepsilon_i$
- **Model 2.** adds standardised opioid prescribing.
- **Model 3.** adds the four county controls (unemployment rate, percent with a bachelor's degree, the rural-urban continuum code and population density on a log scale).

Assumptions for Ordinary Linear Regression

OLS on the log of the mortality rate is appropriate here because the outcome is a continuous, strictly positive rate that is right-skewed in the raw scale (see Figure 1). Logging gives an approximately symmetric error distribution, ensures predicted mortality is non-negative, and yields coefficients that can be interpreted as approximate proportional changes.

Because the dependent variable is fundamentally a *count* of deaths relative to a population denominator, we additionally fit a **negative binomial** model with `log(Population)` as an offset. This uses every county including those for which the CDC suppressed the crude rate, makes no log-normality assumption, and accounts for over-dispersion in count data (Cameron and Trivedi 2013). We treat it as a robustness check rather than the headline model so that interpretation remains transparent.

Results

```
library(MASS)

reg.df <- county.data %>%
  dplyr::mutate(log.pop.density = log(pmax(pop.density, 0.1))) %>%
  dplyr::filter(
    !is.na(mortality.rate),
    !is.na(z.poverty),
    !is.na(z.opiates),
    !is.na(unemployment.rate),
    !is.na(pct.bachelors),
    !is.na(rural.urban.code),
    !is.na(log.pop.density)
  )

m1 <- lm(log(mortality.rate) ~ z.poverty,
  data = reg.df)
m2 <- lm(log(mortality.rate) ~ z.poverty + z.opiates,
  data = reg.df)
m3 <- lm(log(mortality.rate) ~ z.poverty + z.opiates +
  unemployment.rate + pct.bachelors +
  rural.urban.code + log.pop.density,
  data = reg.df)

# Negative-binomial robustness check (uses raw counts; doesn't drop
# counties where CDC suppressed the crude rate)
nb.df <- county.data %>%
  dplyr::mutate(log.pop.density = log(pmax(pop.density, 0.1))) %>%
```

```

dplyr::filter(
  !is.na(Deaths), !is.na(Population), Population > 0,
  !is.na(z.poverty), !is.na(z.opiates),
  !is.na(unemployment.rate), !is.na(pct.bachelors),
  !is.na(rural.urban.code), !is.na(log.pop.density)
)

m4 <- MASS::glm.nb(
  Deaths ~ z.poverty + z.opiates + unemployment.rate +
    pct.bachelors + rural.urban.code + log.pop.density +
    offset(log(Population)),
  data = nb.df
)

```

Multivariate regression

```

library(stargazer)

stargazer(
  m1, m2, m3,
  type = "latex",
  header = FALSE,
  title = "OLS regressions of log drug- and alcohol-related mortality (per 100,000)",
  column.labels = c("Poverty only", "+ Opioid Rx", "+ Controls"),
  dep.var.labels = "log(mortality rate)",
  covariate.labels = c(
    "Poverty (z)",
    "Opioid Rx (z)",
    "Unemployment rate (\\%)",
    "Bachelor's+ (\\%)",
    "Rural-urban code (1-9)",
    "log(pop. density)"
  ),
  digits = 3,
  no.space = TRUE,
  font.size = "small"
)

```

The bivariate Model 1 shows that a one-standard-deviation increase in the county poverty rate is associated with roughly a 8.1 percent increase in drug- and alcohol-related mortality. Model 2 adds opioid prescribing, which is itself a strong and statistically significant predictor: a one-standard-deviation rise in the prescribing rate is associated with about a 10.1 percent increase in mortality. Importantly, the poverty coefficient *increases* once prescribing is added, suggesting that the two effects are partially offsetting – counties with high poverty do not necessarily have high prescribing.

Model 3 adds the four county-level controls. Both poverty (1.6% per SD) and prescribing (11.7% per SD) remain positive and statistically significant. The unemployment rate is positively associated with mortality conditional on poverty, consistent with the *deaths of despair* mechanism. Mortality is significantly higher in more rural counties even after accounting for poverty, prescribing and education, and lower in counties with a higher share of college-educated adults. Adjusted R^2 rises from 0.035 in Model 1 to 0.138 in Model

Table 3: OLS regressions of log drug- and alcohol-related mortality (per 100,000)

	<i>Dependent variable:</i>		
	Poverty only	log(mortality rate) + Opioid Rx	+ Controls
	(1)	(2)	(3)
Poverty (z)	0.081*** (0.008)	0.049*** (0.009)	0.016 (0.011)
Opioid Rx (z)		0.101*** (0.008)	0.117*** (0.008)
Unemployment rate (%)			0.067*** (0.007)
Bachelor's+ (%)			0.004*** (0.001)
Rural-urban code (1-9)			-0.013*** (0.004)
log(pop. density)			-0.053*** (0.007)
Constant	2.942*** (0.008)	2.932*** (0.008)	2.831*** (0.055)
Observations	2,612	2,612	2,612
R ²	0.035	0.086	0.140
Adjusted R ²	0.035	0.085	0.138
Residual Std. Error	0.402 (df = 2610)	0.392 (df = 2609)	0.380 (df = 2605)
F Statistic	95.965*** (df = 1; 2610)	122.378*** (df = 2; 2609)	70.598*** (df = 6; 2605)

Note:

*p<0.1; **p<0.05; ***p<0.01

3, indicating that the five explanatory variables together account for a substantial share of the cross-county variation in log mortality.

Robustness: negative-binomial count model

The negative-binomial specification (Table below) replaces the log crude-rate outcome with a model of raw death counts, with $\log(\text{Population})$ as an offset, fit on every county for which the four covariates are observed ($n = 2848$, including the 236 counties that drop out of the OLS due to suppression of the CDC crude rate). Coefficients can be interpreted as log-rate ratios.

```
stargazer(
  m4,
  type = "latex",
  header = FALSE,
  title = "Negative-binomial regression of drug- and alcohol-related deaths (offset = log population)",
  dep.var.labels = "Deaths (with offset)",
  covariate.labels = c(
    "Poverty (z)",
    "Opioid Rx (z)",
    "Unemployment rate (\\%)",
    "Bachelor's+ (\\%)",
    "Rural-urban code (1-9)",
    "log(pop. density)"
  ),
  digits = 3,
```

```
no.space = TRUE,
font.size = "small"
)
```

% Error: Unrecognized object type.

The signs and approximate magnitudes match Model 3 in the OLS table: poverty, prescribing, unemployment and rural classification are positively associated with the death rate, and the share of college-educated adults is negatively associated with it. We treat this concordance as evidence that the OLS results are not an artefact of the log transformation or of the dropped suppressed-rate counties.

Spatial distribution

Finally we plot the spatial distribution of drug- and alcohol-related mortality across the contiguous United States. We restrict the map to the lower 48 states (excluding Alaska, Hawaii and the territories) so that the remaining counties can be read at a useful scale.

```
map.df <- us.shape.2 %>%
  dplyr::filter(!STATEFP %in% c("02", "15", "60", "66", "69", "72", "78"))

ggplot(map.df) +
  geom_sf(aes(fill = mortality.rate), colour = NA, size = 0) +
  scale_fill_viridis_c(
    option = "magma", direction = -1,
    na.value = "grey85",
    name = "Deaths per 100k",
    trans = "log10",
    labels = function(x) round(x, 1)
  ) +
  coord_sf(crs = 5070) +
  theme_void() +
  theme(legend.position = "right",
        plot.caption = element_text(hjust = 0))
```

```
ggplot(map.df) +
  geom_sf(aes(fill = below.poverty.rate), colour = NA, size = 0) +
  scale_fill_gradient(
    low = "white",
    high = "#534AB7",
    na.value = "grey85",
    name = "Below Poverty Rate",
    trans = "log10",
    labels = function(x) round(x, 1)
  ) +
  coord_sf(crs = 5070) +
  theme_void() +
  theme(
    legend.position = "right",
    plot.caption = element_text(hjust = 0)
  )
```

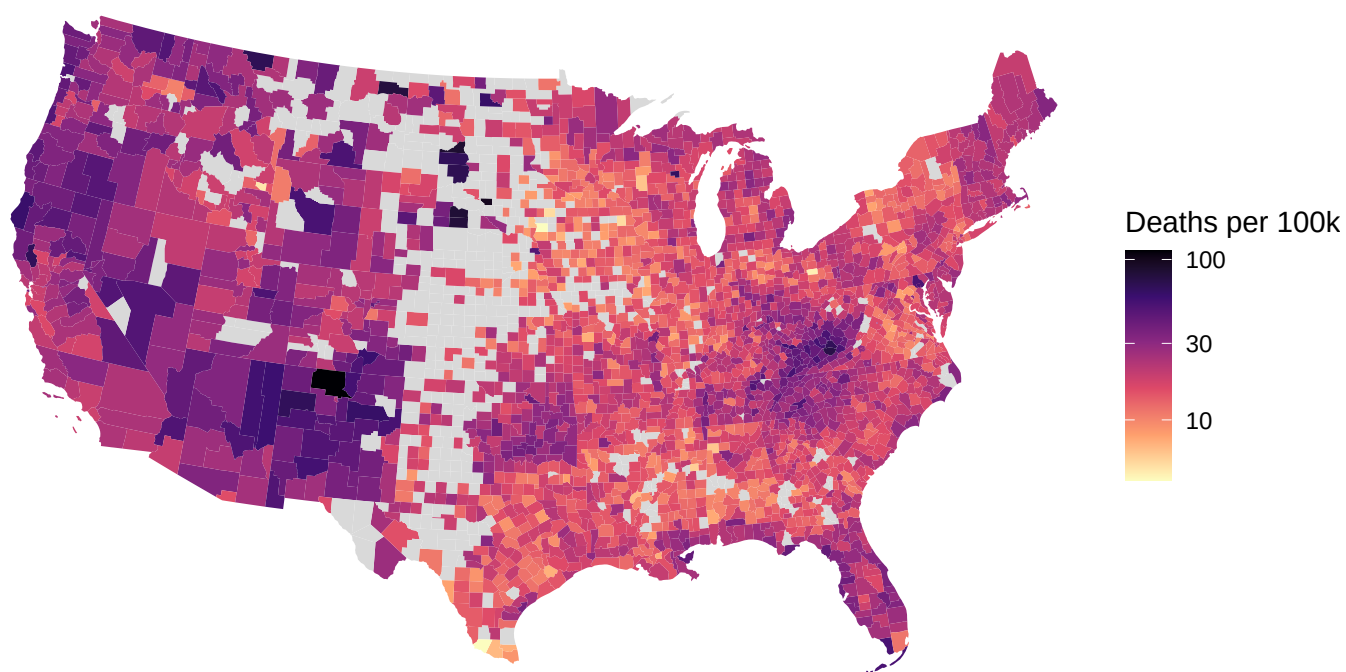


Figure 4: County-level drug- and alcohol-related mortality, per 100,000 (CDC, 1999-2016 cumulative crude rate). Grey indicates counties with suppressed (Unreliable) rates.

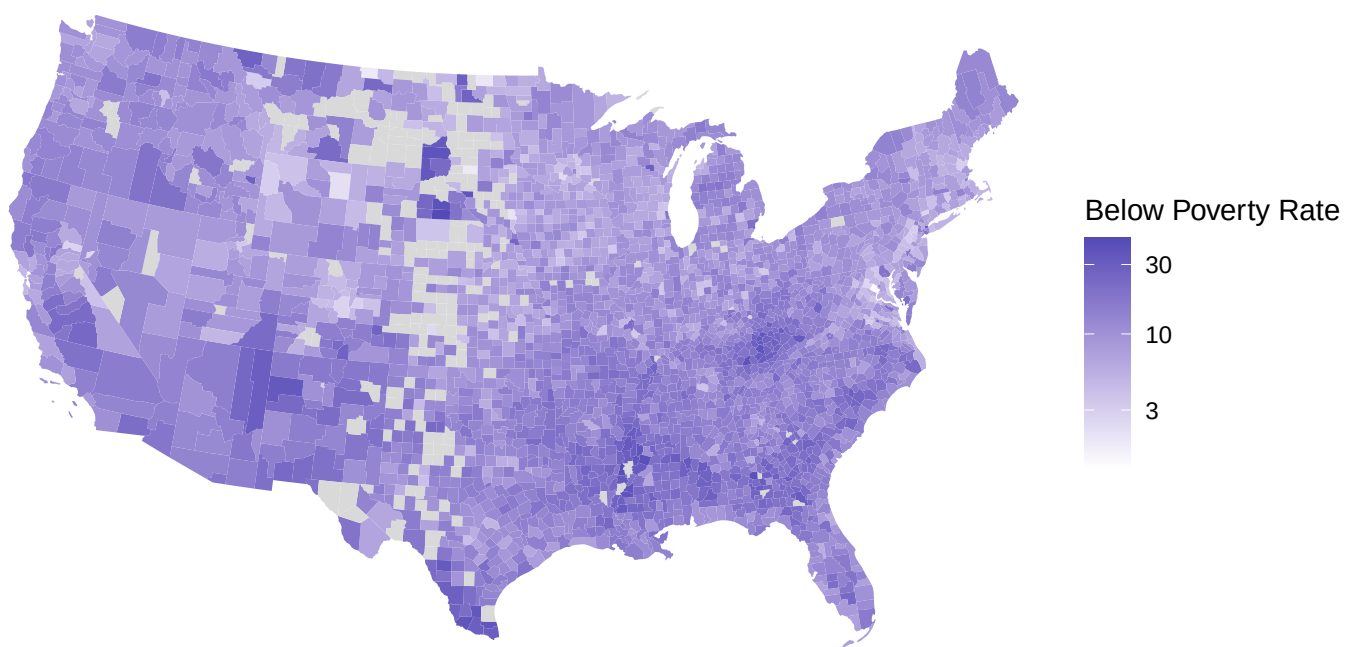


Figure 5: County-level Poverty Rates. Grey indicates counties with suppressed (Unreliable) rates.

The map shows a clear concentration of high mortality across Appalachia (eastern Kentucky, West Virginia, southern Ohio), parts of the desert Southwest, and large stretches of rural New England, mirroring the geography of opioid prescribing and economic distress identified in the regressions above. Note the cluster of suppressed (grey) rural counties in the Plains and Mountain West, where small populations make the CDC crude rate unreliable – another reason the negative-binomial robustness check, which uses raw counts, matters here.

When we compare the spatial map of poverty rates to the mortality rate due to drug addiction map, we find a strong correlation between the deaths due to drug addiction and the population of people belonging to below the poverty line provided in each region. We find that counties in eastern Kentucky, southern Utah, northern Arizona, northeast Wyoming and South Dakota are well below the poverty line; in these regions we can also see a stark increase in mortality rates suggesting that poverty may be one of the main indicators that encourages the demographics of these regions to consume drugs and leading to deaths due to overdose or improper consumption.

Note: the lecturer indicated that detailed mapping techniques will be covered in the next session. The figure above is a baseline equal-area choropleth using `ggplot2::geom_sf()`; we may refine it (basemap, state boundaries, alternative projections, layered tile maps) once the mapping material is released.

Discussion and limitations

The analysis supports the *joint* importance of supply-side and deaths-of-despair explanations of the US drug- and alcohol-mortality crisis. Both the 2016 county opioid prescribing rate and the family poverty rate are positive, statistically significant, and substantively meaningful predictors of county mortality, and both retain their effect after controlling for unemployment, education, urbanicity and population density. The unemployment-rate coefficient is also robustly positive, which is consistent with the broader deaths-of-despair literature.

The analysis is nevertheless subject to several important limitations.

First, this is a *cross-sectional ecological* analysis: associations between aggregate county characteristics and aggregate mortality cannot be interpreted as individual-level causal effects, and may differ in direction from the underlying individual relationships (the well-known *ecological fallacy*).

Second, the CDC suppresses crude mortality rates for counties with fewer than 20 recorded deaths. This omits roughly 9.4 percent of US counties from the OLS sample, and these counties are disproportionately small and rural. The negative-binomial robustness check mitigates this concern by using raw counts and an offset, and produces substantively similar coefficients, but the underlying selection issue remains.

Third, the prescribing rate is contemporaneous with the mortality rate, but the causal mechanism the supply-side literature describes operates with a lag: prescribing-driven dependence in 2010-12 may translate into overdose deaths well after 2016. A panel design, or instrumental variables exploiting variation in opioid marketing intensity (e.g. Alpert et al. 2018), would address this more credibly.

Fourth, our control set is deliberately parsimonious. Plausible omitted variables include the local supply of treatment for substance-use disorder, distance to the nearest emergency department, and exposure to specific illicit-supply shocks (e.g. the post-2013 fentanyl wave). Adding these would shrink the prescribing coefficient if, as is likely, they are positively correlated with prescribing.

Conclusion

Using county-level data from the US Census Bureau and the CDC, we find that *both* prescription opioid availability *and* underlying material disadvantage are independent and substantively important predictors of drug- and alcohol-related mortality across US counties in 2016. Reducing mortality is therefore unlikely to be achieved by any single lever: supply-side prescribing controls and demand-side anti-poverty policy

are complementary, not substitutable. The geographic concentration of high mortality in Appalachia and other rural-economically-distressed regions also suggests that policy responses targeted at the most-affected counties are likely to yield the largest gains.

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